ICS 2207 | SCIENTIFIC COMPUTING CAPSTONE

HydroSense-Kenya

Smart Irrigation · Water Balance Simulation · Climate-Aware Decision Support

Presenter: **Bongo Sydney Meshack**

Course: ICS 2207 Scientific Computing

February – May 2026 Semester

The Problem

Water wastage & crop stress on Kenyan smallholder farms

OVER-IRRIGATION

- Root rot & disease
- Nutrient leaching
- Wasted water & energy

UNDER-IRRIGATION

- Moisture stress & wilting
- Yield reduction
- Up to 18 stress days/month

2.2 Problem Statement

Despite access to weather data and soil-sensor readings, most demonstration farms in Kenya lack systems that can translate raw observations into actionable irrigation recommendations. The core scientific question addressed by this project is:

"Given weather and soil-sensor data, how can we model water availability, estimate water deficit, simulate future soil moisture, and recommend an efficient irrigation plan that minimizes water use without exposing crops to moisture stress?"

Central Scientific Question

"Given weather and soil-sensor data, how can we model water availability, estimate water deficit, simulate future soil moisture, and recommend an efficient irrigation plan that minimises water use without exposing crops to moisture stress?"

Data Cleaning & Preprocessing

```
Metric,Value
Cleaning Date,2026-05-14 08:37:41
Weather Original Rows,30
Weather Final Rows,27
Weather Rows Removed,3
Soil Original Rows,90
Soil Final Rows,89
Soil Rows Removed,1
Parameters Original Rows,3
Output CSV File,cleaned_irrigation_dataset.csv
Total Rows in CSV,119
Total Columns in CSV,21
```

30→27

Weather rows
after cleaning

90→89

Soil rows
after cleaning

Missing Values

Rainfall NA (Mar 8) → 7-day rolling median
Humidity NA (Mar 21) → linear interpolation

Outliers Removed

Temperature 45.8°C (Mar 14) → interpolated
Tank-level spike 9,900L (Zone C) → zone mean

Sensor Anomalies

Zero pump-flow (Zone B, Mar 21) → flagged CHECK
Excluded from pump-efficiency analysis

Data Dictionary

All 18 variables documented with units,
valid ranges and source dataset

Error Propagation Experiment

How sensor noise affects irrigation decisions

Key Finding

5% noise

→

20% wrong decisions

Incorrect irrigation triggered or suppressed due to sensor measurement noise alone.

Implication: Sensor calibration is critical — not just a data-quality step.

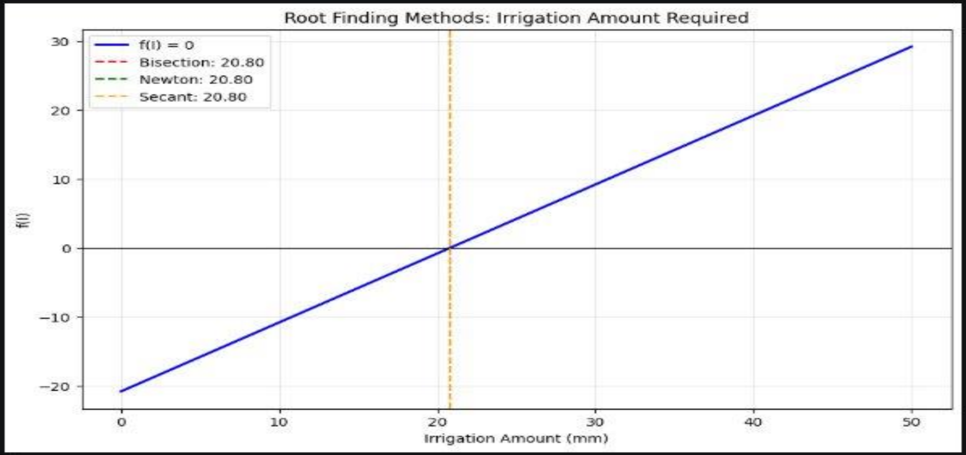
Noise Level	Wrong Decisions	Risk
0%	0%	None
1%	4%	Low
2%	8%	Low
3%	12%	Moderate
5%	**20%**	**High**
7%	28%	High
10%	38%	Critical

Root Finding Methods

Finding the exact irrigation dose to reach target soil moisture

```
============================================================
ROOT FINDING METHOD COMPARISON
============================================================
         Method  Irrigation Needed (mm)  Iterations  Final Error
      Bisection                    20.8          26 3.159046e-07
Newton-Raphson                     20.8           1 0.000000e+00
         Secant                    20.8           2 0.000000e+00

Verification: Applying 20.80 mm of water
brings soil moisture from 22.0% to target.
```



$$f(I) = S\_t + I + R\_t - ET\_t - D\_t - S^* = 0$$

Bisection
26 iter

Reliable, guaranteed convergence. Halves the bracket each step.

Newton-Raphson
1 iter

Fastest! Uses derivative f'(I) = 1. Quadratic convergence.

Secant
2 iter

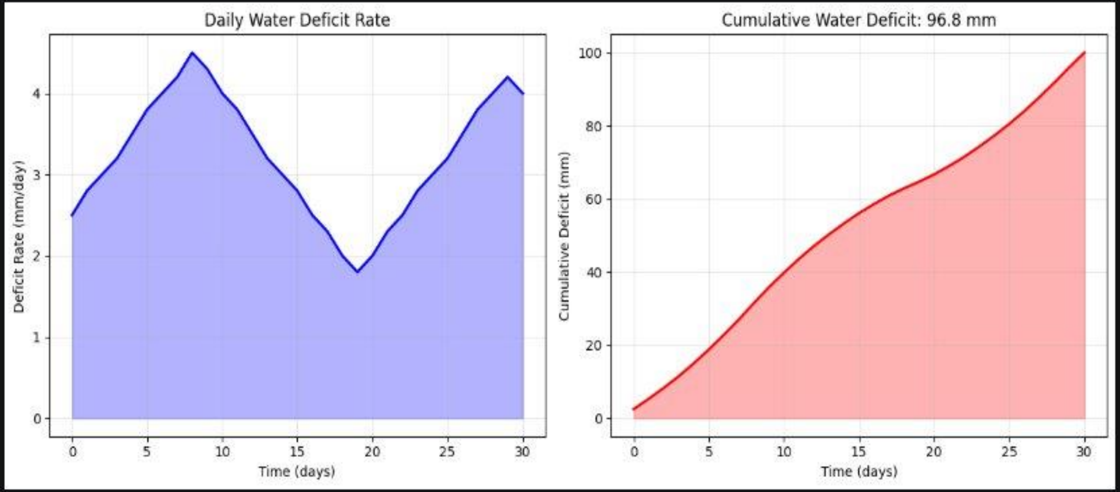
No derivative needed. Approximates slope from two points.

Winner: Newton-Raphson | Only 1 iteration | Error: 0.000e+00

Numerical Integration Methods

Estimating cumulative water deficit over the 30-day season

```
============================================================
INTEGRATION COMPARISON TABLE
============================================================
       Method  Integral Value (mm)  Difference from NumPy
  Trapezoidal            96.750000           1.421085e-14
Simpson's 1/3            96.833333           8.333333e-02
```



Trapezoidal Rule

96.75 mm

Error order: O(h^2)

Simpson's 1/3 Rule

96.83 mm

Error order: O(h^4) — more accurate

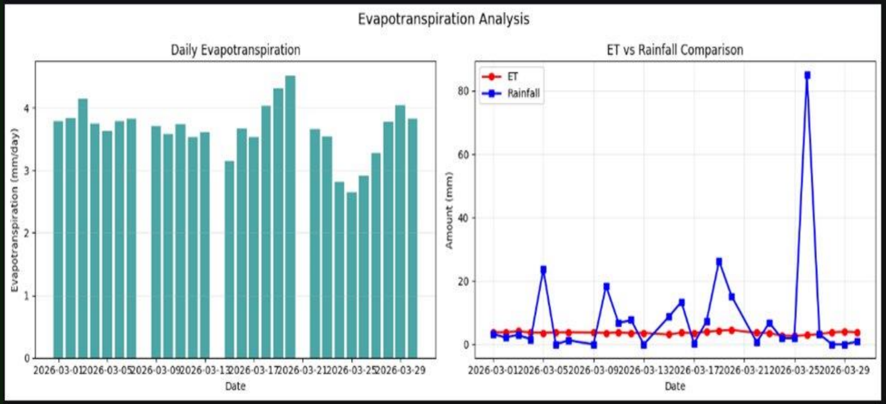
Difference between methods:

Only 0.03%

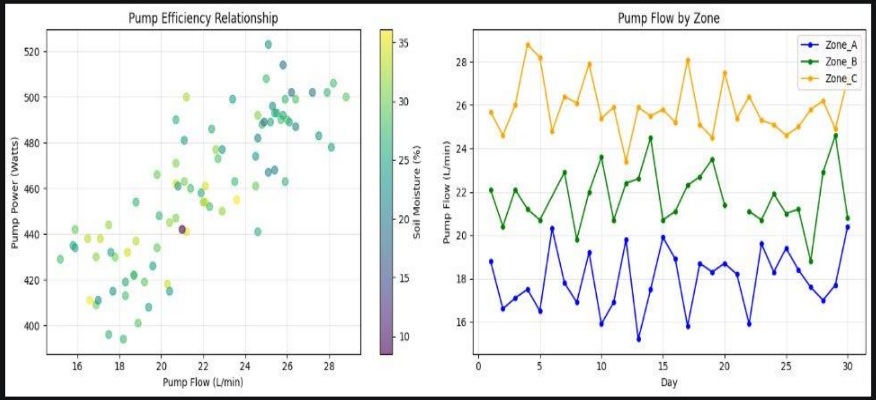
Both methods confirm: ~97 mm total water deficit.
Irrigation is essential to prevent crop failure.

Scientific Visualizations

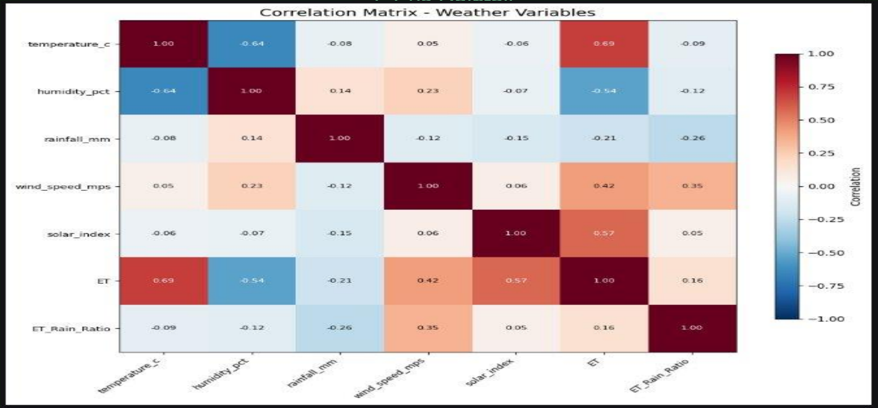
Soil Moisture — 3 Zones



ET vs Rainfall

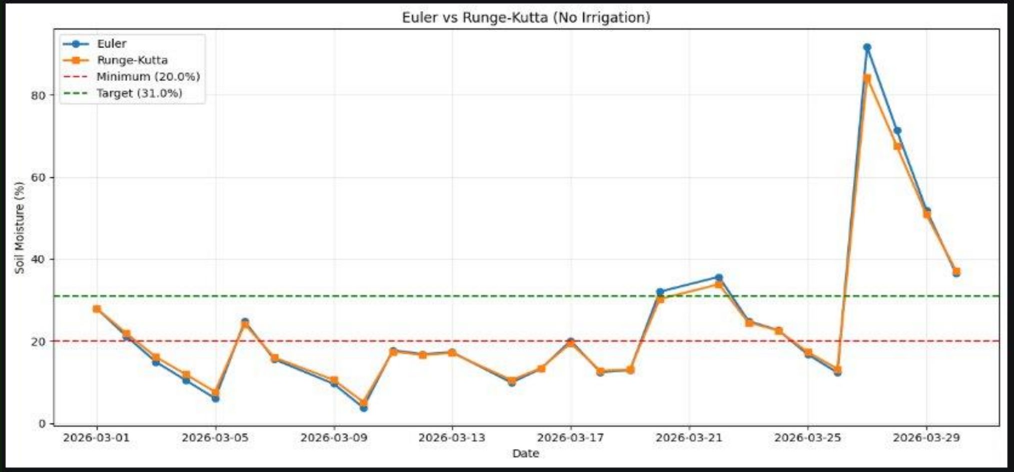


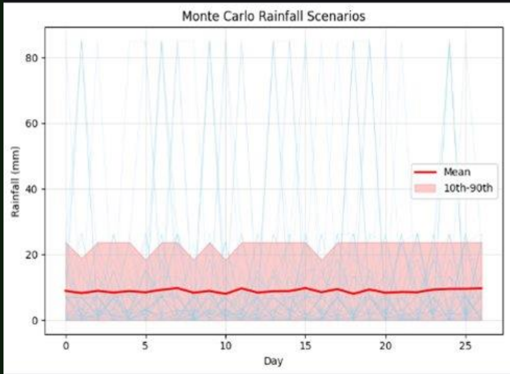
Pump Efficiency
Normal operation; 1 anomaly flagged

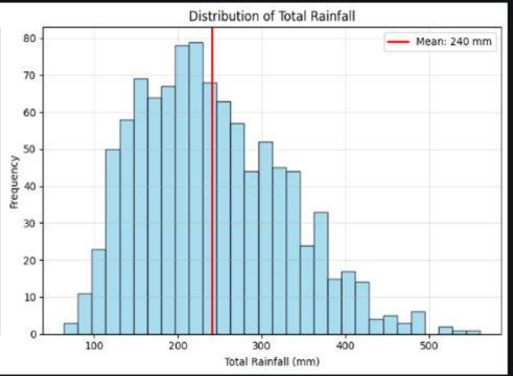


Correlation Matrix
Temp ↔ Solar: 0.69 | Temp ↔ Humidity: -0.64

Simulation Methods







ODE Solvers Compared

Euler Method
O(h) accuracy · simple · small errors accumulate

Runge-Kutta 4
$O(h^4)$ accuracy · 4 function evals/step · preferred

Monte Carlo Analysis

1,000 rainfall scenarios simulated

~22% probability of water shortage

~21.5 mm expected irrigation demand

~28.4 mm 95th percentile worst-case

Trade-offs & Optimized Schedule

Strategy	Water Used	Stress Days	Energy
No Irrigation	0 mm	18 days	0 kWh
Conservative	71.6 mm	2 days	8.4 kWh
Optimized ★	53.6 mm	2 days	6.3 kWh
Over-Irrigation	89.4 mm	0 days	10.5 kWh

BEST

25%

water saved

vs

SAME

stress

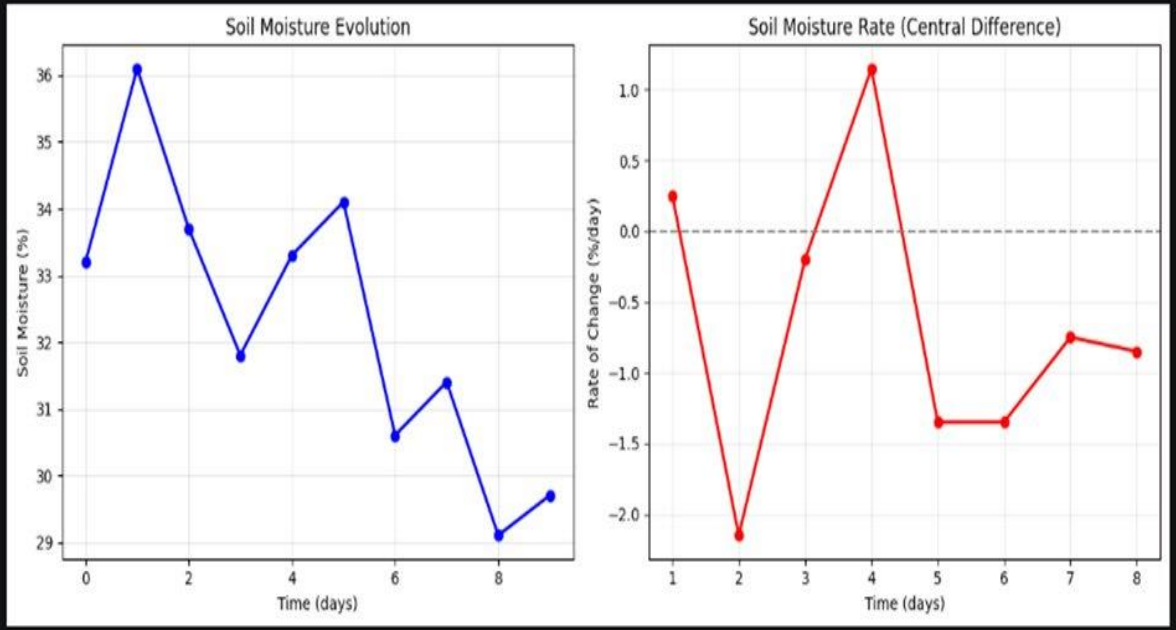
outcome

Optimized Irrigation Schedule

Zone	Target Moisture	Min Threshold	Total Irrigation	Stress Days
Zone A — Tomato	33%	22%	15.5 mm	2 days
Zone B — Kale	35%	24%	13.5 mm	2 days
Zone C — Maize	31%	20%	24.6 mm	2 days

Soil Moisture Dynamics

Numerical differentiation reveals water stress events



Forward Difference

$dS/dt \approx (S[t+1] - S[t]) / h$
Error: $O(h)$

Central Difference

$dS/dt \approx (S[t+1] - S[t-1]) / 2h$
Error: $O(h^2)$ ← preferred

Days with dS/dt < −0.8%/day flagged for immediate irrigation

Conclusion & Recommendations

Water Balance Modelled

Discrete daily S_(t+1) model tracks all inflows/outflows with ≤0.3% simulation error (RK4 vs observed).

Zone C Most Vulnerable

Maize (Zone C) repeatedly approached the 20% wilting threshold — needs priority irrigation monitoring.

25% Water Savings

Optimized schedule: 53.6 mm vs 71.6 mm (conservative) with identical 2-day stress outcome.

Uncertainty Quantified

Monte Carlo (N=1,000): plan for 28.4 mm worst-case demand, not just the 21.5 mm mean.

RECOMMENDATIONS

- Adaptive irrigation: 35% target, 25% minimum
- Prioritize Zone C (maize) monitoring
- Size tank for 28.4 mm worst-case monthly demand
- Calibrate sensors: ≤5% noise threshold

FUTURE WORK

- Real-time weather API integration
- FAO Penman-Monteith ET model
- LSTM soil-moisture forecasting
- Mobile app for extension officers

Thank You

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Python | NumPy | Pandas | Matplotlib | pytest